Forensic Repair of a Damaged MBR Partition Table

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Tools Used: 010 Editor, FTK Imager

1. Incident Description

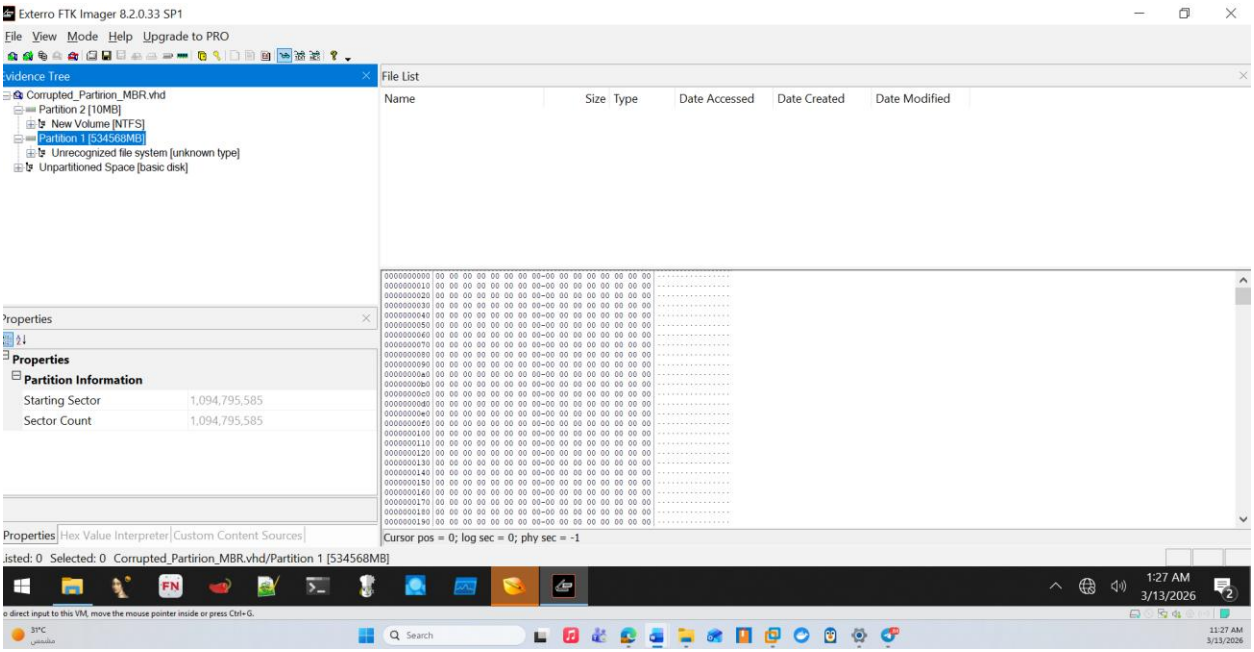
In this scenario, the 16-byte first partition entry inside the Master Partition Table was intentionally overwritten, while the rest of the Master Boot Record remained intact.

This type of corruption prevented the forensic tool from correctly identifying the first partition, even though the NTFS volume itself still existed deeper inside the disk.

```
24 02 C3 49 6E 76 61 6C 69 64 20 70 61 72 74 69  $.Ãinvalid parti
74 69 6F 6E 20 74 61 62 6C 65 00 45 72 72 6F 72  tion table.Error
20 6C 6F 61 64 69 6E 67 20 6F 70 65 72 61 74 69   loading operati
6E 67 20 73 79 73 74 65 6D 00 4D 69 73 73 69 6E  ng system.Missin
67 20 6F 70 65 72 61 74 69 6E 67 20 73 79 73 74  g operating syst
65 6D 00 00 00 63 7B 9A 7B DA 2E CE 00 00 41 41  em...c{š{Ú.Î..AA
41 41 41 41 41 41 41 41 41 41 41 41 41 41 00 8E  AAAAAAAAAAAAAA.Ž
0D 02 07 D4 11 03 80 A0 00 00 00 50 00 00 00 00  ...Ô..€ ...P....
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00 00 00 00 00 00 00 00 00 00 00 00 00 00 55 AA  ..............Uª
```

2. Initial Examination

The disk image was opened in FTK Imager:



This indicated that only the first partition descriptor required manual reconstruction

3. Manual Analysis of the First Partition Entry

The damaged first partition entry inside the Master Partition Table was manually reconstructed based on the standard 16-byte MBR partition entry structure

Partition entry reconstruction logic:

Offset	Size (Bytes)	Description	Value Used
0	1	Boot Indicator	00 (Non-bootable) or 08 (bootable)
1	3	Starting CHS	00 00 00 (no longer used)
4	1	Partition Type	07 (NTFS) or 0B (FAT32)
5	3	Ending CHS	00 00 00 (no longer used)
8	4	Relative Sector (LBA Start)	calculated manually
12	4	Number of Sectors in Partition	calculated manually

3. Manual Reconstruction of the First Partition Entry

The corrupted first partition entry was manually rebuilt according to the standard 16-byte Master Partition Table structure used in MBR disks.

3.1 Boot Indicator and CHS Fields

The first byte of the partition entry represents the **Boot Indicator**.

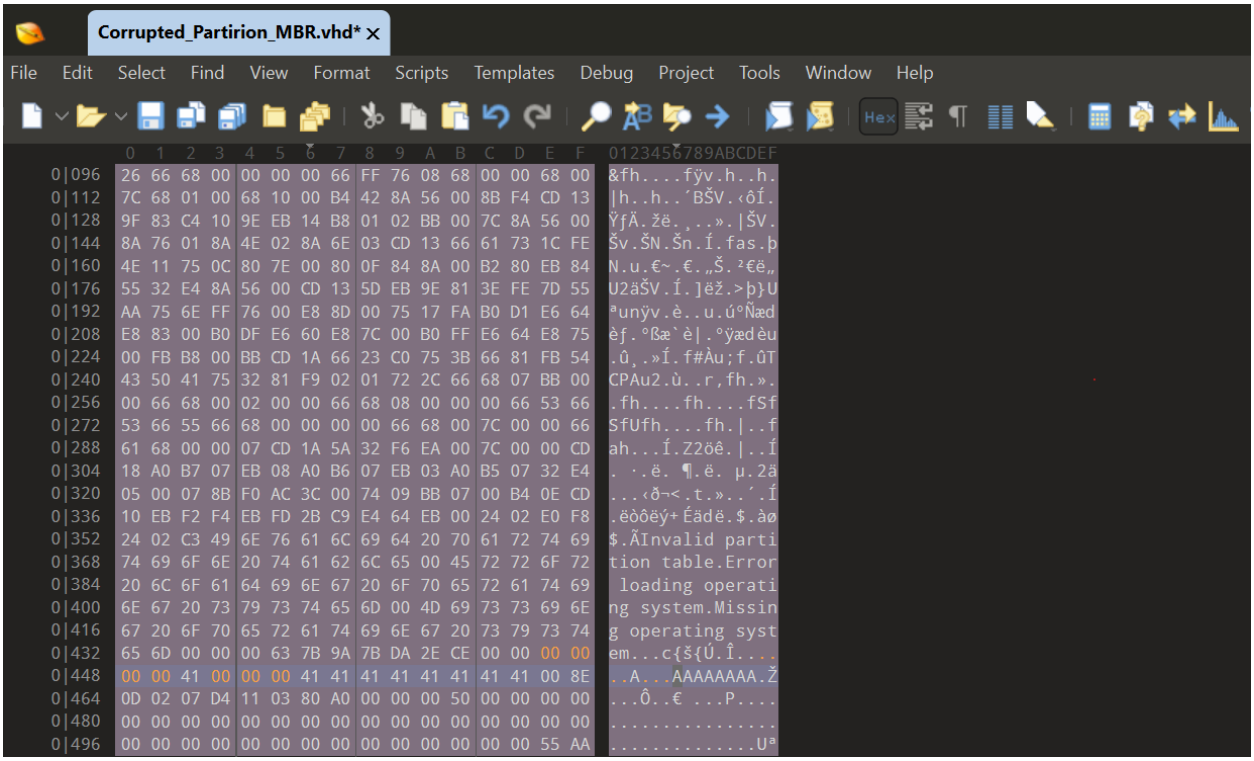
Since the partition in this case was not bootable, the following value was inserted: **00**

The next three bytes represent the **Starting CHS (Cylinder-Head-Sector)** values.

Because CHS addressing is no longer required for modern forensic recovery and LBA values are used instead, these bytes were manually filled with: **00 00 00**

The same approach was applied to the **Ending CHS** field, where the next three bytes were also set to: **00 00 00**

This ensured structural consistency in the partition entry while relying on LBA for actual recovery.



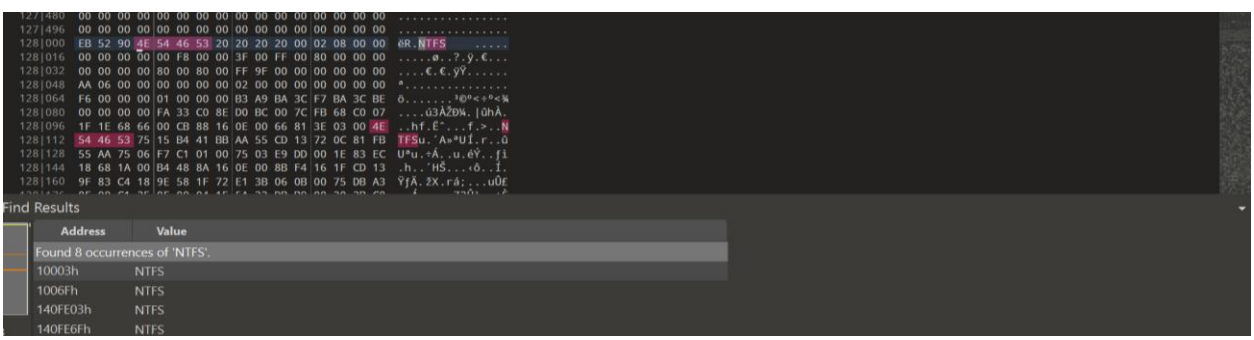
3.2 Determining the Partition Type

To identify the correct file system type, the disk image was searched for the NTFS**.** This confirmed that the surviving volume used the NTFS file system.

Therefore, the partition type byte was assigned: **07**

3.3 Determining the Relative Sector (LBA Start)

To identify the starting sector of the damaged partition, the NTFS signature located inside the surviving volume was used as the reference point.



The search returned **8 occurrences**.

This result is expected because each NTFS partition normally contains:

- one primary Volume Boot Record
- one backup boot sector

Within each of these sectors, the NTFS signature appears twice in nearby positions.

Therefore:

- one NTFS partition produces 4 NTFS occurrences
- two NTFS partitions produce 8 NTFS occurrences

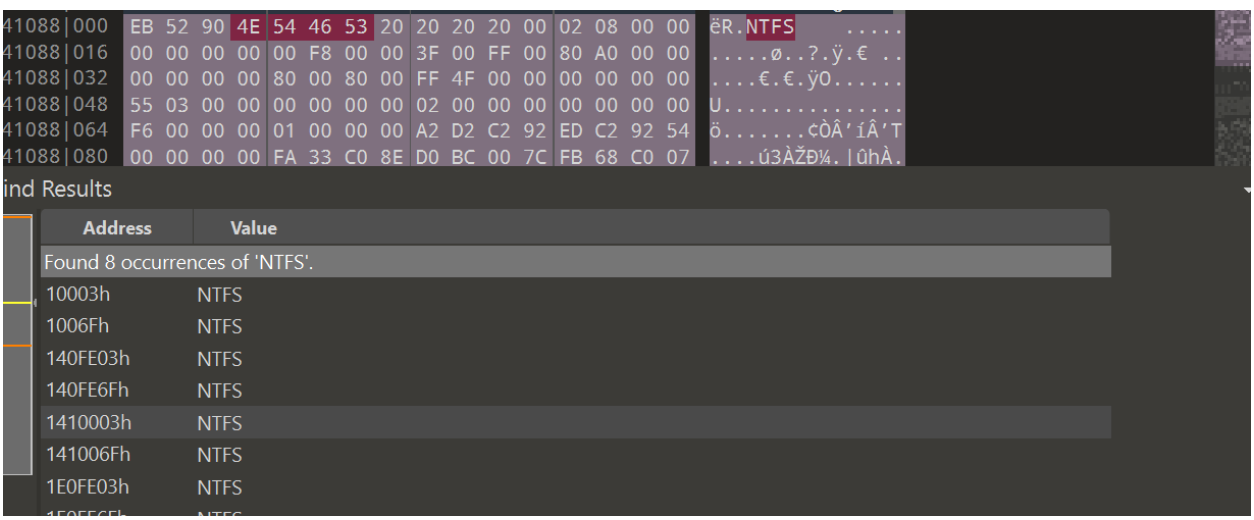
The first NTFS signature was found at sector 128

Relative Sector (LBA Start) = 128

This value was written on the partition table in little-endian format as: **80 00 00 00**

3.4 Determining the Number of Sectors in the Partition

To calculate the total size of the first partition, the starting sector of the second partition was identified.



The second partition begins at: **41088**

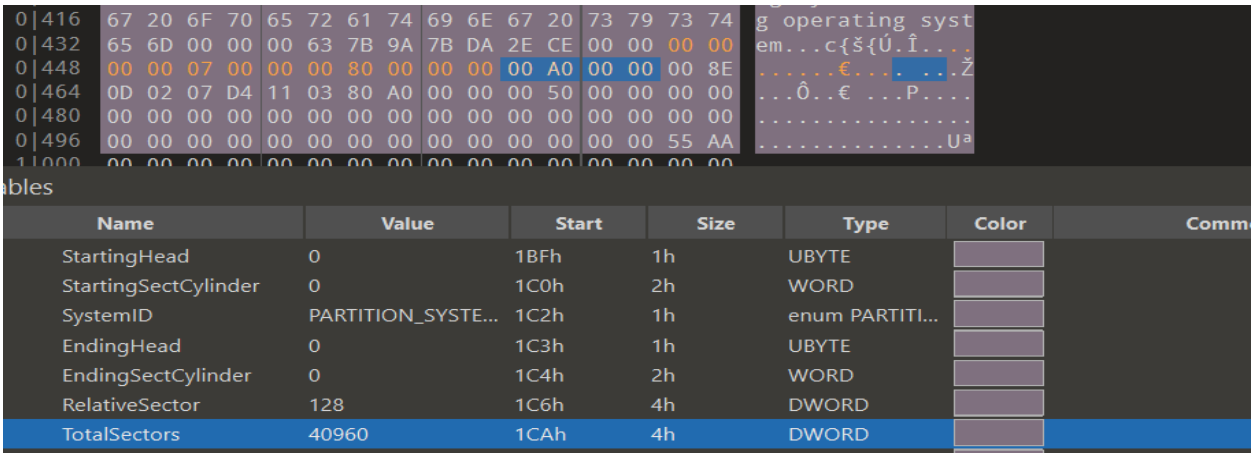
Because the first partition begins at sector 128, the total number of sectors was calculated as: 41088-128=40960

Thus: **Number of Sectors = 40960**

Convert to hexadecimal: **A000**

Little-endian representation: **00 A0 00 00**

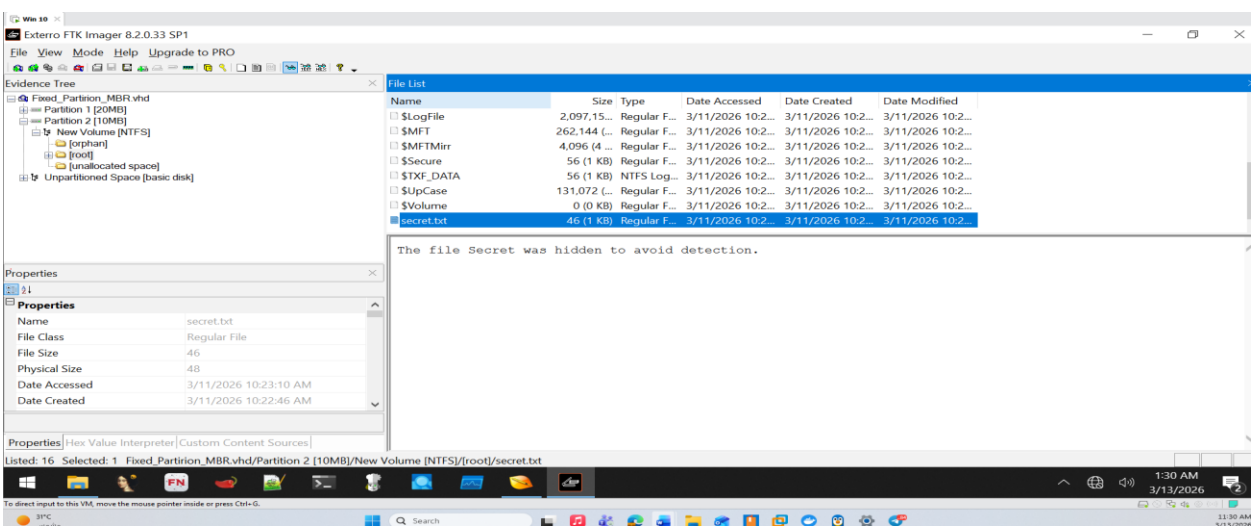
This value was inserted into the **Number of Sectors field** of the partition entry.



4. Verification of the Recovered Disk in FTK Imager

After manually reconstructing the corrupted partition entry, the modified disk image was reopened using FTK Imager to verify whether the partition could be recognized correctly.

The forensic tool successfully interpreted the repaired partition table and displayed the previously inaccessible NTFS partition as a valid file system.



Recovered contents inside the partition included:

- secret.txt

The successful appearance of files inside the repaired partition demonstrated that the NTFS volume itself had remained intact and that only the partition reference inside the MBR had been damaged.